

1. FILE / SIGNAL INFORMATION

File Name	synthetic_BPSK_1000kHz_SNR20dB.wav	Modulation	BPSK
File Format	WAV Audio / IQ (WAV)	Modulation Confidence	100.00%
File Size	89,876 bytes	Occupied Bandwidth	173.828 kHz (173.828.1 Hz)
Signal Duration	17.41 ms (0.01741 s)	Carrier Freq Offset (CFO)	-4.88 Hz
Sample Rate (Fs)	1,000 MHz (1,000,000.0 Hz)	Symbol Rate	Not available
Center Freq Offset	-4.88 Hz	Samples Per Symbol	Not available
Total Samples	17,408	Frame Sync Status	Unlocked / Direct

2. DETECTED SIGNAL PARAMETERS

3. MODULATION CLASSIFICATION

Modulation	Probability	Distribution
BPSK	100.00%	
QPSK	0.00%	
GMSK	0.00%	

4. TIME-DOMAIN WAVEFORM

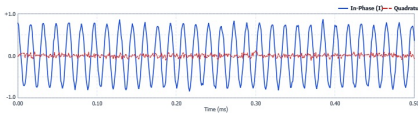


Figure 1: Time-Domain Waveform (In-Phase I & Quadrature Q Components)

5. FREQUENCY SPECTRUM



Figure 2: Power Spectral Density (PSD) and Occupied Bandwidth

6. SPECTROGRAM / WATERFALL

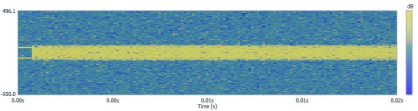


Figure 3: Time-Frequency STFT Spectrogram Waterfall

7. CONSTELLATION DIAGRAM

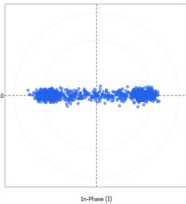


Figure 4: In-Phase vs Quadrature (IQ) Constellation Diagram

8. SYNCHRONIZATION / FRAME ANALYSIS

Header Offset	0 bits	Payload Offset	0 bits
Correlation Sync Match	Not available	Sync Pattern Locked	Direct framing
Phase Inversion Resolved	No (Direct polarity)	Demodulated Bits Count	128 bits

9. BITSTREAM / DECODING

Total Decoded Bits	128 bits	Decoded Bytes	16 bytes
Bit Density (1s / 0s)	64 (50.0%) / 64 (50.0%)	ASCII Text Preview	BPSK TEST PAYLOAD
Bitstream Sample	01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101		

Decoded Bytes Hex Dump:

```
000000 42 58 53 48 28 54 45 53 54 28 58 41 59 4C 4F 41 |BPSK TEST PAYLOAD|
000008 44 |0|
```

10. FEC / DE-INTERLEAVER ANALYSIS

Component	Detected Configuration	Status / Confidence
De-interleaver	None	Bypassed
Forward Error Correction (FEC)	None	Uncoded
Channel Decoder	Automated Multi-Scheme Pipeline	16 bytes recovered

11. FINAL ANALYSIS SUMMARY

RF capture file 'synthetic_BPSK_1000kHz_SNR20dB.wav' was processed, containing 17,408 samples recorded at a sample rate of 1,000 MHz (1,000,000.0 Hz) (effective duration: 17.41 ms (0.01741 s)). The neural modulation classifier identified the transmission format as BPSK with a classification confidence of 100.00%. Spectral analysis determined an occupied bandwidth of 173.828 kHz (173.828.1 Hz) with a carrier frequency offset of -4.88 Hz. Demodulation recovered 128 raw bit transitions. Channel decoding detected direct (no) interleaving, and raw uncoded bit framing, successfully assembling 16 decoded payload bytes.